5. System Architecture

The Process Change Impact Simulator will use a modular web-based architecture. The application will be developed using Python and Flask in Visual Studio Code. SQLite will store process models and change scenarios. NetworkX will support dependency analysis, while Pandas and NumPy will help process and compare simulation data. Matplotlib and Plotly will be used to present graphs and simulation results.

The major components and their responsibilities are explained below.

- User Interface: Process analysts can create or import workflows, select proposed changes, run impact analysis, and review results through web pages built with HTML and CSS.

- Process Model Manager: This component manages process definitions, activities, decision conditions, and dependencies. It allows users to create, edit, save, and import process models.

- Database Layer: SQLite stores process information, activity details, dependencies, change scenarios, and analysis results. JSON files will support process import and export.

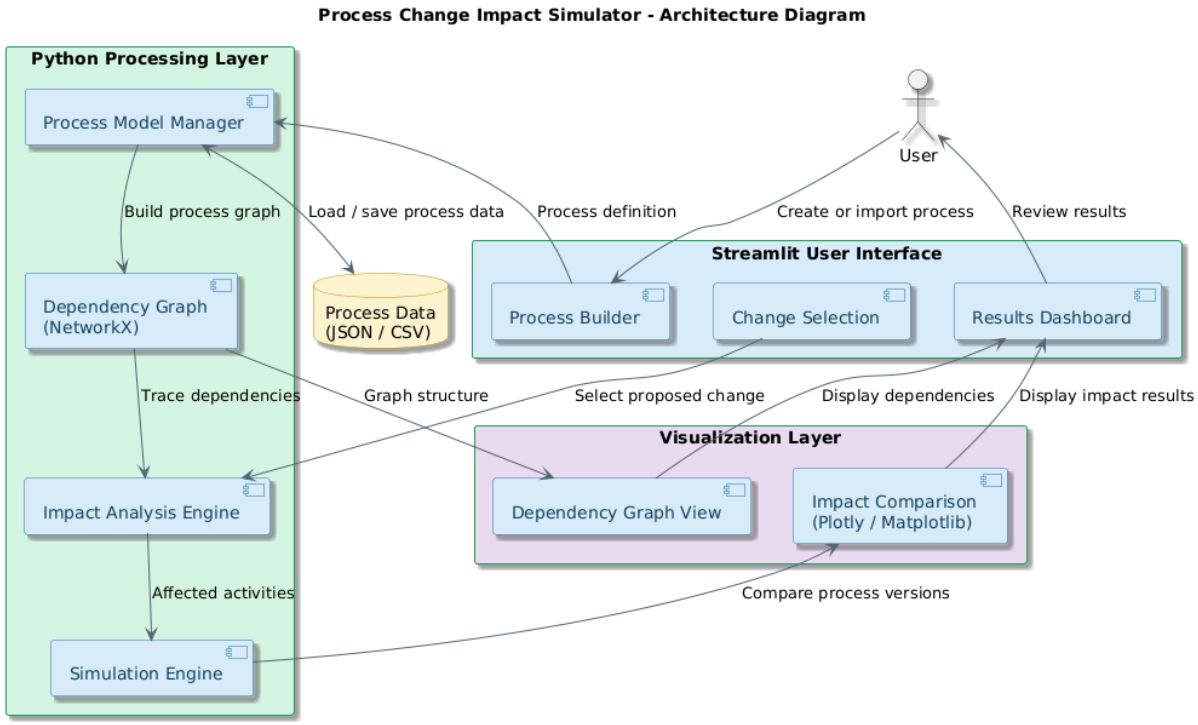
- Dependency Graph Builder: NetworkX represents activities as graph nodes and dependencies as directed edges. This component checks process connections and identifies circular dependencies.

- Impact Analysis Engine: This component traces downstream activities from a selected change and identifies connected tasks, approvals, and outputs that may be affected.

- Simulation Engine: This component runs the original and modified process scenarios and compares execution paths, skipped activities, newly required steps, and final outputs.

- Visualization and Results Dashboard: Plotly and Matplotlib display dependency graphs and comparisons. Process managers can review affected activities and export analysis reports.

System interaction: Users submit process data through the web interface. Flask passes requests to the appropriate processing component, which retrieves or saves information in SQLite. NetworkX performs dependency analysis, and the simulation engine compares process versions. The results are returned to the interface for visualization and review.


Process Change Impact Simulator - Architecture Diagram
Python Processing Layer
Process Model Manager
Build process graph
Load / save process data
Process definition
Dependency Graph (NetworkX)
Process Data (JSON / CSV)
Trace dependencies
Graph structure
Impact Analysis Engine
Affected activities
Simulation Engine
User
Create or import process
Review results
Streamlit User Interface
Process Builder
Change Selection
Results Dashboard
Select proposed change
Display dependencies
Display impact results
Visualization Layer
Dependency Graph View
Impact Comparison (Plotly / Matplotlib)
Compare process versions